Quantum Rope Perturbation Theory

The One-Loop Fluctuation Determinant — and Why It Falsifies the One-Loop Mass Mechanism

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper reports the semiclassical (one-loop) perturbation theory of fluctuations around a rope soliton, and it reports a NEGATIVE result, plainly. The quantity that would control a fluctuation-generated mass is the regularised log-determinant of the second-variation operator, $\ln[\det'(-\nabla^2+V)/\det(-\nabla^2)]$. Computed for the physical (positive, localised) second-variation potential, this ratio is an O(1) number with NO negative modes — the solver returns −1.29 for a representative well. The electron scale, by contrast, would require a log-determinant of about 108. The one-loop fluctuation determinant is therefore roughly two orders of magnitude too small, and of the wrong character, to generate the lepton mass scale. The one-loop mass mechanism is falsified by its own computation. We report this as a finding. All numbers are outputs of the rope_solver spectrum computation in this package (log_det_ratio, poschl_teller_bound_states, required_log_det_for_electron), not inserted constants.

The negative result, stated plainly

The one-loop fluctuation log-determinant around a rope soliton is O(1) (computed ≈ −1.29) with no negative modes. The electron scale requires ≈ 108. The one-loop determinant is ~80× too small and lacks the instability (negative mode) a mass-generating mechanism would need.

CONCLUSION: quantum fluctuations at one loop do NOT generate the lepton mass scale in the rope model. This is a falsification of the one-loop mass mechanism, kept as a finding, not removed.

1. One-Loop Perturbation Theory Around a Soliton

A classical rope soliton contributes to quantum amplitudes with a weight set, at one loop, by the determinant of the operator governing small fluctuations around it — the second variation of the action, $M = -\nabla^2 + V$, where V is the (positive, localised) second-variation potential. The physically meaningful, regularised quantity is the ratio to the free determinant,

$$\Delta = \ln[\, \det'(-\nabla^2 + V) / \det(-\nabla^2)\,] \quad (1.1)$$

computed from the low-lying spectra of the two operators on a common three-dimensional grid. If fluctuations were to generate a large mass scale, Δ would need to be large (and typically the spectrum would carry a negative mode signalling an instability).

2. The Computation and Its Result

Evaluated for the physical second-variation potential, the solver returns a small, negative, O(1) value with no negative modes:

Quantity	Computed	What a mass mechanism needs
One-loop log-det Δ	≈ −1.29 (O(1))	large (≈ 108 for the electron)
Negative modes	0	≥ 1 (instability) typically required
Ratio to electron requirement	~1/80	order unity

The gap is not marginal. The one-loop determinant is about two orders of magnitude below the value the electron scale requires, and it lacks the negative mode that a genuine mass-generating instability would exhibit. No reasonable adjustment within one-loop perturbation theory closes a factor of eighty.

3. What This Falsifies, and What Remains

The result is specific and worth stating precisely.

3.1 Falsified

- The one-loop fluctuation mass mechanism: the hypothesis that quantum fluctuations around a rope soliton, at one loop, generate the lepton mass scale. The computed determinant is far too small and structurally wrong (no negative mode). This is a clean falsification by the programme's own computation.

3.2 What remains open

- Whether some non-perturbative or higher-order mechanism could supply the required scale is not addressed here and is not claimed. The negative result is specific to one-loop perturbation theory.
- The electron log-determinant requirement (≈ 108) remains, as reported in the knot-fluctuation paper, a TARGET that no computed rope spectrum has been shown to meet. This paper closes off the one-loop route to that target.

Why report it. A natural and attractive idea — that the lepton mass emerges from quantum fluctuations around the knot — is tested directly and fails at one loop, by a large and unambiguous margin. Recording that failure prevents the corpus from resting on a mechanism its own computation rules out, and narrows the search: whatever sets the lepton mass in the rope model, it is not the one-loop fluctuation determinant.

4. Status of Each Result

Result	Status
One-loop log-det Δ ≈ −1.29, O(1), no negative modes	Derived — direct spectral computation
Electron requires Δ ≈ 108	Modeled — target (see knot-fluctuation paper)
One-loop fluctuation mass mechanism	Failed — falsified by ~80× gap; kept as finding
Non-perturbative mass mechanism	Open — not addressed, not claimed

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Conclusion

Quantum rope perturbation theory, carried to one loop, returns a negative result: the fluctuation log-determinant around a rope soliton is an O(1) quantity (computed ≈ −1.29) with no negative modes, while the electron scale would require roughly 108. The one-loop mechanism for generating the lepton mass from fluctuations is thereby falsified by the model's own computation, by about two orders of magnitude. In keeping with the programme's practice, the failure is reported as a finding and retained: it rules out an attractive but incorrect mechanism, and it sharpens the open question of what, if anything, sets the lepton mass scale in the rope model — which is not, the computation shows, one-loop fluctuations.

Registered Claims in This Paper
Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

• **QB-004 [Failed]:** One-loop fluctuation mass mechanism

POSTSCRIPT — THE ELECTROMAGNETIC ARC (registered 2026-08-09, claims EM-RECON-024 through 028)
Five chartered commissions, each with bars locked before computation, closed the carrier question this paper's channel-map left open and rebuilt the electromagnetic sector from registered mechanics. (1) THE DIRECTION (EM-RECON-024): under the operational definition the static sector already owned — E is force per winding — the angular-averaged linear response of the azimuth-blind contact form to a passing wave is a scalar times the identity: the force is parallel to the carrier's transverse center-line displacement. The necessity clause unified two registered kills:

no center-line displacement, no force, no polarization — the bare screw's F = 0 IS the state-count kill seen from the force side.

(2) THE CARRIER (EM-RECON-025): the collective transverse displacement of the fully-dynamical mesh is a symmetry-protected gapless Goldstone pair — the exact two-strand computation puts the O(g) crossing mass entirely on the relative (optical) branch while the center-of-mass branch runs at omega^2 = (T0/mu) q^2, gapless at every crossing strength. The earlier mass kill was a frozen-background artifact, and the escape is located in the operator, not argued. Costs on the face: the coverage-vs-dispersive fork becomes mandatory, and the bending condition B <= T0 a^2/12 stands as the live falsifier.

(3) THE SIGN (EM-RECON-026): the q-linear force is derived from the momentum-flux (Blasius) integral — F = rho (v_rel x Gamma) with Gamma = q kappa_0 by the pi_1 = Z winding-charge identification — and the electric field is identified: E = rho kappa_0 (v x zhat). The static sign rule (like windings repel) is reproduced from momentum conservation, independently confirming the energy-minimization route; and the relative-velocity split yields the exact Lorentz structure F = q(E + v_d x B'), electricity and magnetism as the two halves of one Magnus formula.

(4) THE NORMALIZATION (EM-RECON-027): two independent routes — wave-energy bookkeeping under the registered field-strain bridge, and static Gauss consistency — fix the identical identity kappa_0^2 eps0 rho = 1, whence kappa_0 = c/sqrt(eps0 SIGMA): the circulation quantum is the bridge's one constant in electromagnetic clothes, and the Coulomb law emerges with no residual factor. The registered Schwinger-form bound on SIGMA propagates to kappa_0 <= ~26-50 SI; a SIGMA measurement fixes every magnitude in the sector simultaneously.

(5) THE MAGNETIC SECTOR (EM-RECON-028): the ether-drag worry — the honest way the Lorentz split could have died — cancels as a computed identity: the Magnus axis is the winding's own line direction, so the net drag on a closed winding is the closed integral of the tangent, identically zero. The same topology that quantizes charge protects free motion. Faraday applied to the derived E forces mu_0 = 1/(eps0 c^2) from the carrier's own dispersion, and the structure lands on the registered magnetic claims (EM-009/012) as agreement between independent routes.

What remains, named: full covariance of the defect sector, beyond-plane-wave bookkeeping, and SIGMA's value — the single gate on which every magnitude hangs. Five commissions, zero free parameters introduced.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.